dynamic visibility supports faster decision-making and adaptive traffic regulation, improving urban mobility efficiency.

In urban planning, UAV-based 3D mapping provides accurate spatial data for land-use analysis, construction planning, zoning, and environmental impact assessment. Drones help planners visualize terrain conditions, assess building density, identify heat islands, and monitor green space changes. This contributes to sustainable city design and supports data-driven policy-making.

By linking drones to smart city IoT networks, city authorities can integrate UAV-collected data into public transportation planning, emergency services coordination, and environmental monitoring platforms, making cities more responsive and interconnected.

8.7.3 Urban air mobility

Urban air mobility (UAM) represents the next frontier in drone-enabled smart cities, where passenger-carrying autonomous aerial vehicles will transport people across urban regions, bypassing congested road networks. Companies and governments are developing electric vertical take-off and landing (eVTOL) air taxis that can operate from small skyports integrated into city infrastructure. These systems promise to significantly reduce travel times for inter-city commutes and emergency medical evacuations.

Pilot programs in cities such as Dubai, Singapore, Paris, and Los Angeles demonstrate growing interest in UAM. However, large-scale adoption requires advancements in airspace integration, noise mitigation, battery performance, public safety regulations, and societal acceptance. Air traffic management systems must evolve to support low-altitude aerial corridors shared by drones and eVTOL vehicles.

Despite these challenges, UAM holds the potential to redefine the future of transportation. It aligns with sustainability goals by supporting electric